

B.E. Electrical Engineering
Second Year Second Semester

Power Supply Systems

Part-II Complete Solutions

2024 Question Paper

Academic Article Format

Simple English, SN Ma'am Style, Exam-Oriented

Answering Style

Definition → Diagram → Assumption → Formula → Derivation → Stepwise
Numerical → Boxed Final Answer

Prepared for clear semester-writing practice

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Note

This document solves all Part-II questions of the 2024 Power Supply Systems paper. The writing style is kept simple and stepwise. Diagrams are drawn with enough spacing so that labels do not collapse or go outside the page margin.

Question 1 Group A, Question 1**Question**

- (i) Explain why loop arrangement is better than radial arrangement in distribution systems.
- (ii) What do you understand by feeders, distributors and service mains?
- (iii) Derive the generalized expression for voltage drop and point of minimum potential in a two-wire distributor having uniformly distributed loading and fed at one end.

1.1 (i) Loop arrangement and radial arrangement**Simple definition**

A **radial distribution system** supplies power from one end only. A **loop or ring distribution system** forms a closed path and can supply a load point from two sides.

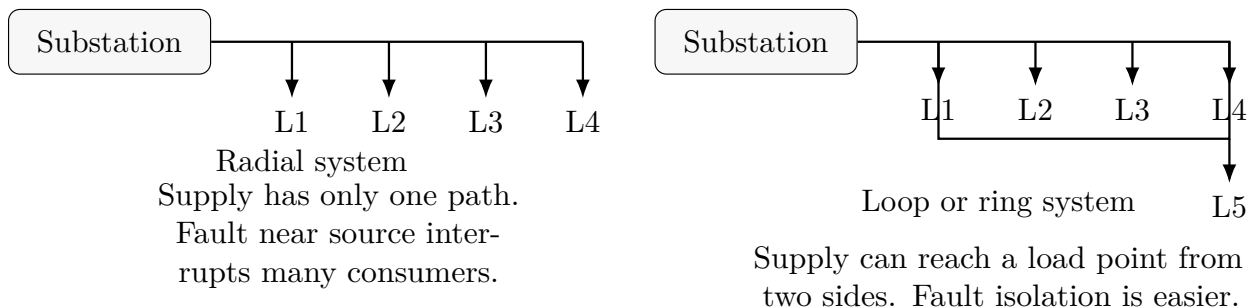


Figure 1: Radial and loop distribution systems

Why loop arrangement is better

1. **Better continuity of supply:** If one section is faulty, supply can be maintained from the other side.
2. **Better voltage regulation:** The voltage drop is smaller because current is shared by two paths.
3. **Less copper for same performance:** Since current is divided, section currents are lower, so conductor size may be reduced.
4. **Easy fault isolation:** The faulty section can be isolated by switches without switching off the whole area.

5. **Suitable for important areas:** It is better for town distribution, industrial loads and important consumers.

Final Answer

A loop arrangement is better because it gives better reliability, lower voltage drop, better voltage regulation and easier fault isolation compared with a radial arrangement.

1.2 (ii) Feeder, distributor and service mains

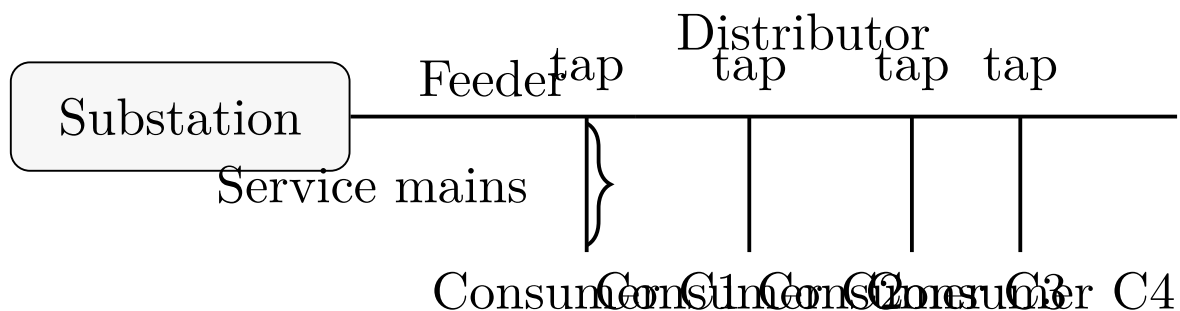


Figure 2: Feeder, distributor and service mains

Term	Simple meaning
Feeder	It carries power from substation to the distribution area. Normally no tapping is taken from a feeder. It is designed mainly from current-carrying capacity.
Distributor	It is the conductor from which several tappings are taken for consumers. It is designed mainly from voltage-drop point of view.
Service mains	It connects the consumer terminals to the distributor. It is the final small connection to the consumer.

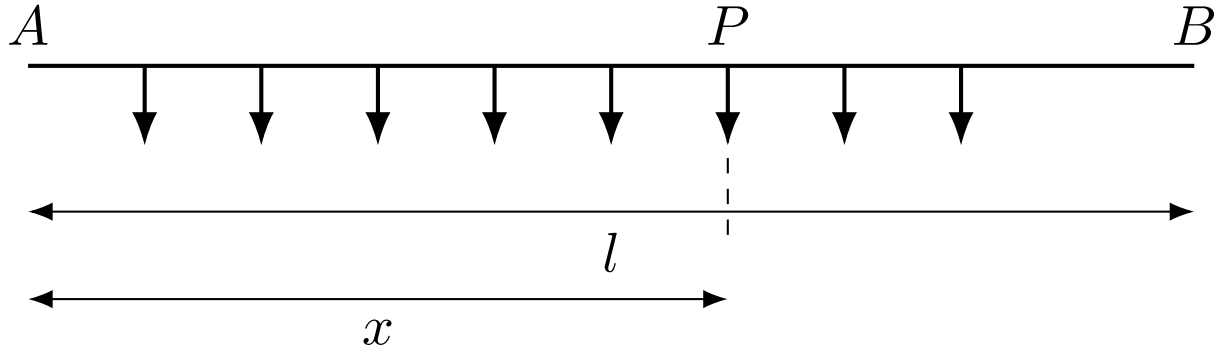
SN Ma'am Method

In exam, write the difference clearly: feeder is current-capacity based, distributor is voltage-drop based, and service mains are consumer connection conductors.

1.3 (iii) Uniformly loaded two-wire DC distributor fed at one end

Diagram and notation

Let the distributor length be l metre. It is fed at end A and uniformly loaded at i ampere per metre. Let loop resistance, including go and return, be r ohm per metre.



Uniform load = i A/m, loop resistance = $r \Omega/\text{m}$

Figure 3: Two-wire distributor fed at one end with uniform loading

Current at a section

At a distance s from the feeding end A , the remaining load beyond that section is $i(l - s)$. Therefore, current in the small section ds is

$$I_s = i(l - s).$$

Voltage drop across small length ds is

$$dv = I_s \times r ds = i(l - s)r ds.$$

Voltage drop up to a point P at distance x from A is

$$v_x = \int_0^x i(l - s)r ds.$$

$$v_x = ir \left(lx - \frac{x^2}{2} \right)$$

At the far end, $x = l$,

$$v_{\max} = ir \left(l^2 - \frac{l^2}{2} \right) = \frac{ir l^2}{2}.$$

If total current $I = il$ and total loop resistance $R = rl$, then

$$v_{\max} = \frac{IR}{2}.$$

Point of minimum potential

Voltage at distance x is

$$V_x = V_A - ir \left(lx - \frac{x^2}{2} \right).$$

For minimum potential, voltage drop should be maximum. The drop is maximum when $x = l$. Hence, the minimum potential occurs at the far end B .

Final Answer

For a uniformly loaded two-wire distributor fed at one end,

$$v_x = ir \left(lx - \frac{x^2}{2} \right), \quad v_{\max} = \frac{ir l^2}{2} = \frac{IR}{2}.$$

The point of minimum potential is the farthest end from the feeding point.

Question 2 Group A, Question 2

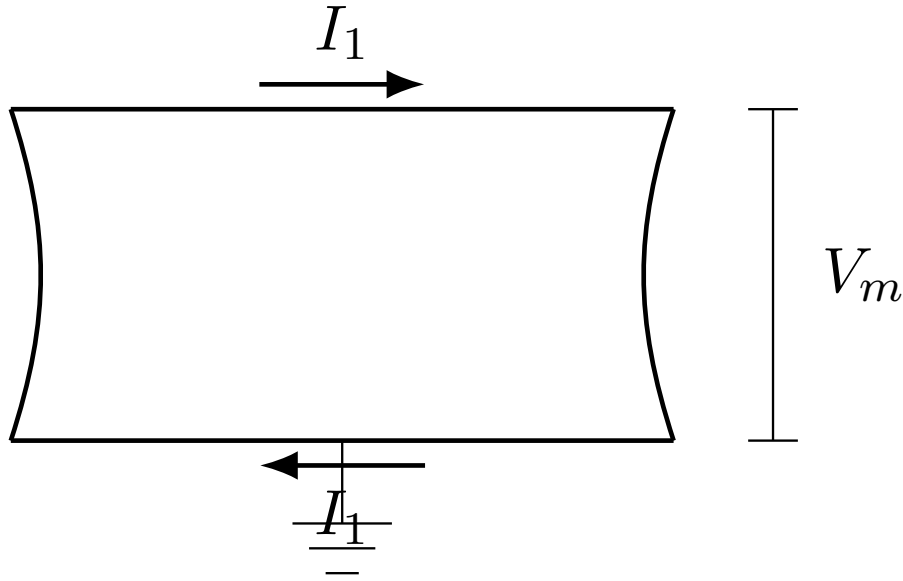
Question

- (i) Compare the volume of conductor material required for DC two-wire system, one conductor earthed, with AC two-phase four-wire system in underground lines. State the assumptions.
- (ii) A single-phase AC system supplies a certain power. If this system is converted to a three-phase AC system by running a third similar copper conductor, calculate the additional percentage load that can be supplied for the same voltage between conductors, same power factor and same percentage copper loss.

2.1 (i) DC two-wire one conductor earthed and AC two-phase four-wire underground system

Common assumptions

1. Same power P is transmitted.
2. Same length of transmission line l is used.
3. Same power loss W is allowed.
4. Same conductor material is used, so resistivity ρ is same.
5. For underground cable, maximum voltage between any two conductors is same and equal to V_m .
6. Power factor for AC system is $\cos \phi$.

Case 1: DC two-wire system, one conductor earthed

one conductor earthed

Figure 4: DC two-wire system with one conductor earthed

For DC system,

$$P = V_m I_1 \quad \Rightarrow \quad I_1 = \frac{P}{V_m}.$$

Let area of each conductor be a_1 . Resistance of each conductor is

$$R_1 = \frac{\rho l}{a_1}.$$

Total line loss is

$$W = 2I_1^2 R_1 = 2 \left(\frac{P}{V_m} \right)^2 \frac{\rho l}{a_1}.$$

So,

$$a_1 = \frac{2P^2 \rho l}{W V_m^2}.$$

Volume of conductor material is

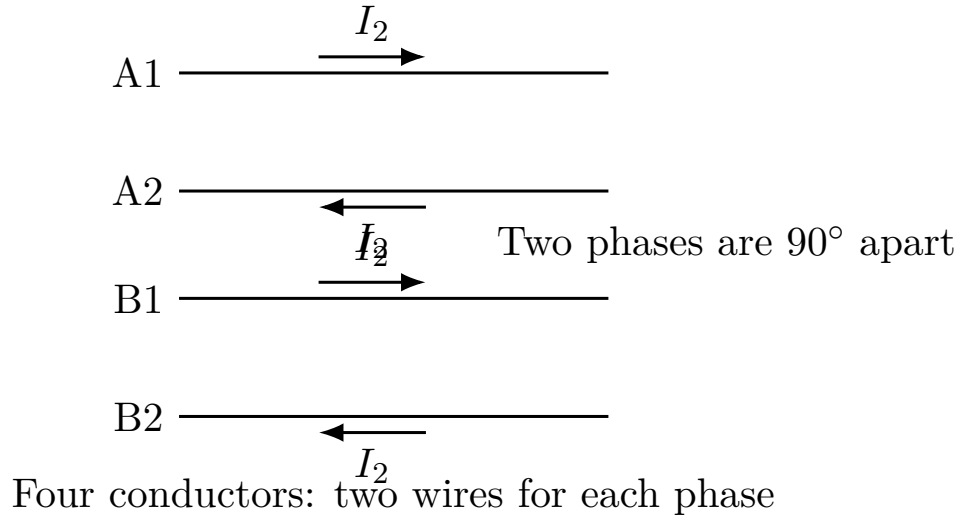
$$V_{dc} = 2a_1 l = \frac{4P^2 \rho l^2}{W V_m^2}.$$

Let

$$\boxed{K = \frac{4P^2 \rho l^2}{W V_m^2}}.$$

Therefore,

$$\boxed{V_{dc} = K}.$$

Case 2: AC two-phase four-wire underground system**Figure 5:** AC two-phase four-wire system

For underground system, maximum voltage between any two conductors is V_m . In a two-phase four-wire system the two phase voltages are at 90° . Hence, voltage of each phase circuit is

$$V_{ph} = \frac{V_m}{\sqrt{2}}.$$

Power transmitted is

$$P = 2V_{ph}I_2 \cos \phi = 2 \left(\frac{V_m}{\sqrt{2}} \right) I_2 \cos \phi = \sqrt{2}V_m I_2 \cos \phi.$$

Thus,

$$I_2 = \frac{P}{\sqrt{2}V_m \cos \phi}.$$

Let area of each conductor be a_2 . Total copper loss is

$$W = 4I_2^2 \frac{\rho l}{a_2} = 4 \left(\frac{P}{\sqrt{2}V_m \cos \phi} \right)^2 \frac{\rho l}{a_2}.$$

$$W = \frac{2P^2 \rho l}{V_m^2 \cos^2 \phi a_2}.$$

So,

$$a_2 = \frac{2P^2 \rho l}{W V_m^2 \cos^2 \phi}.$$

Volume of conductor material is

$$V_{2\phi,4w} = 4a_2 l = \frac{8P^2 \rho l^2}{W V_m^2 \cos^2 \phi}.$$

Since

$$K = \frac{4P^2 \rho l^2}{WV_m^2},$$

we get

$$V_{2\phi,4w} = \frac{2K}{\cos^2 \phi}.$$

Final Answer

$$\frac{V_{2\phi,4w}}{V_{dc}} = \frac{2}{\cos^2 \phi}.$$

So the AC two-phase four-wire underground system requires $\frac{2}{\cos^2 \phi}$ times the conductor material required by the DC two-wire system with one conductor earthed.

2.2 (ii) Single-phase AC changed to three-phase AC by adding one similar conductor

Given condition

- Same voltage between conductors.
- Same power factor.
- Same conductor resistance per wire.
- Same percentage copper loss.

Single-phase two-wire system

Let the load current be I_1 and voltage between conductors be V .

$$P_1 = VI_1 \cos \phi.$$

Copper loss is

$$W_1 = 2I_1^2 R.$$

Percentage copper loss is

$$\%W_1 = \frac{2I_1^2 R}{VI_1 \cos \phi} \times 100 = \frac{200I_1 R}{V \cos \phi}.$$

Three-phase three-wire system

Let the line current be I_2 .

$$P_2 = \sqrt{3}VI_2 \cos \phi.$$

Copper loss is

$$W_2 = 3I_2^2 R.$$

Percentage copper loss is

$$\%W_2 = \frac{3I_2^2 R}{\sqrt{3}V I_2 \cos \phi} \times 100 = \frac{100\sqrt{3}I_2 R}{V \cos \phi}.$$

Since percentage copper losses are same,

$$\frac{200I_1 R}{V \cos \phi} = \frac{100\sqrt{3}I_2 R}{V \cos \phi}.$$

$$2I_1 = \sqrt{3}I_2 \Rightarrow I_2 = \frac{2I_1}{\sqrt{3}}.$$

Now,

$$P_2 = \sqrt{3}V \left(\frac{2I_1}{\sqrt{3}} \right) \cos \phi = 2V I_1 \cos \phi = 2P_1.$$

Percentage additional load is

$$\frac{P_2 - P_1}{P_1} \times 100 = \frac{2P_1 - P_1}{P_1} \times 100 = 100\%.$$

Final Answer

$$\boxed{\text{Additional load supplied} = 100\%}$$

Therefore, by adding a third similar conductor and converting single-phase AC to three-phase AC, the load can be doubled for the same voltage, same power factor and same percentage copper loss.

Question 3 Group A, Question 3

Question

- (a) Prove Kelvin's law for economic size of conductor.
 (b) A two-conductor cable is required to supply a constant load of 200 A throughout the year. The cost of the cable is Rs. $(50a + 25)$ per metre, where a is the area of each conductor in cm^2 . The line is in operation for 250 days in a year. Find the most economical cross-section of the conductor assuming interest and depreciation of 10% per annum and energy cost of 5 paise per unit. Specific resistivity of conductor is $1.85 \mu\Omega \text{ m}$.

3.1 (a) Kelvin's law

Simple statement

Kelvin's law says that the most economical size of a conductor is obtained when:

$$\text{Annual variable capital charge} = \text{Annual cost of energy lost}$$

Derivation

Let a be the cross-sectional area of the conductor.

Total annual cost has two parts:

$$C = \text{annual capital charge} + \text{annual cost of energy loss.}$$

1. Annual capital charge

$$C_1 = P_1 + P_2 a.$$

Here P_1 is the constant part and $P_2 a$ is the variable part depending on conductor area.

2. Annual cost of energy loss

Resistance is inversely proportional to area:

$$R = \frac{\rho l}{a}.$$

So energy loss cost is inversely proportional to a :

$$C_2 = \frac{P_3}{a}.$$

Therefore,

$$C = P_1 + P_2 a + \frac{P_3}{a}.$$

For minimum annual cost,

$$\frac{dC}{da} = 0.$$

$$\frac{d}{da} \left(P_1 + P_2 a + \frac{P_3}{a} \right) = 0.$$

$$P_2 - \frac{P_3}{a^2} = 0.$$

$$P_2 = \frac{P_3}{a^2}.$$

Multiplying by a ,

$$P_2 a = \frac{P_3}{a}.$$

Here $P_2 a$ is the annual variable capital charge and $\frac{P_3}{a}$ is the annual cost of energy wasted.

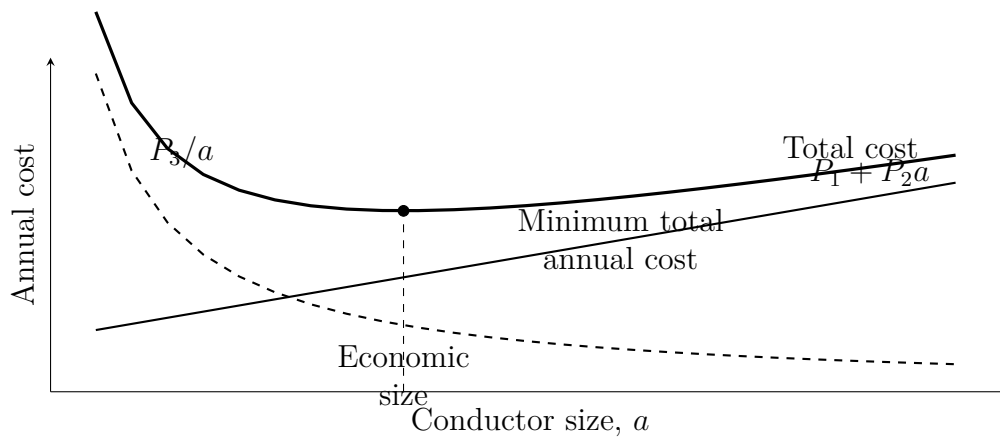


Figure 6: Kelvin's law cost curve

Final Answer

Kelvin's law is proved as:

$$\text{Variable part of annual capital charge} = \text{Annual cost of energy wasted}$$

This gives the most economical conductor size.

3.2 (b) Numerical: most economical conductor area

Given data

$$I = 200 \text{ A}, \quad t = 250 \times 24 = 6000 \text{ h/year.}$$

$$\text{Cost of cable per metre} = 50a + 25 \text{ Rs/m.}$$

Variable capital cost per metre is Rs. $50a$.

Annual interest and depreciation is 10%, so annual variable charge is

$$C_{cap} = 0.10 \times 50a = 5a \text{ Rs/m/year.}$$

Energy cost is 5 paise/unit:

$$5 \text{ paise} = 0.05 \text{ Rs/kWh.}$$

Resistance per metre

Area of each conductor is $a \text{ cm}^2$:

$$A = a \times 10^{-4} \text{ m}^2.$$

As printed in the question,

$$\rho = 1.85 \times 10^{-6} \Omega \text{m}.$$

Resistance of each conductor per metre is

$$R = \frac{\rho l}{A} = \frac{1.85 \times 10^{-6} \times 1}{a \times 10^{-4}} = \frac{0.0185}{a} \Omega.$$

For two conductors, power loss per metre is

$$P_{\text{loss}} = 2I^2 R = 2(200)^2 \frac{0.0185}{a} = \frac{1480}{a} \text{ W}.$$

Energy lost per year per metre is

$$E = \frac{1480}{a} \times \frac{6000}{1000} = \frac{8880}{a} \text{ kWh}.$$

Annual cost of energy lost is

$$C_{\text{loss}} = \frac{8880}{a} \times 0.05 = \frac{444}{a} \text{ Rs/m/year}.$$

By Kelvin's law,

Annual variable capital charge = Annual cost of energy lost.

$$5a = \frac{444}{a}.$$

$$a^2 = \frac{444}{5} = 88.8.$$

$$a = \sqrt{88.8} = 9.42 \text{ cm}^2.$$

Final Answer

Taking the printed resistivity as $1.85 \times 10^{-6} \Omega \text{m}$,

$$\boxed{a = 9.42 \text{ cm}^2}$$

Therefore, the most economical cross-section of each conductor is 9.42 cm^2 .

Note

In many textbooks copper resistivity is written as $1.85 \mu\Omega \text{ cm}$, not $1.85 \mu\Omega \text{ m}$. If the intended unit was $1.85 \mu\Omega \text{ cm}$, then the answer becomes approximately 0.942 cm^2 . In the main solution

above, the unit printed in the question has been followed.

Question 4 Group A, Question 4

Question

(i) A DC ring main distributor ABCDA is fed from point A from a 250 V supply. The resistances, including go and return, are: $AB = 0.02\Omega$, $BC = 0.018\Omega$, $CD = 0.025\Omega$, $DA = 0.02\Omega$. Loads are 150 A at B, 300 A at C and 250 A at D. Determine voltage at each load point. If A and C are linked through an interconnector of resistance 0.02Ω , determine the new voltage at each load point.

(ii) Show that transmission line efficiency increases with increase in transmission line voltage.

4.1 (i) DC ring main distributor

Step 1: Ring main without interconnector

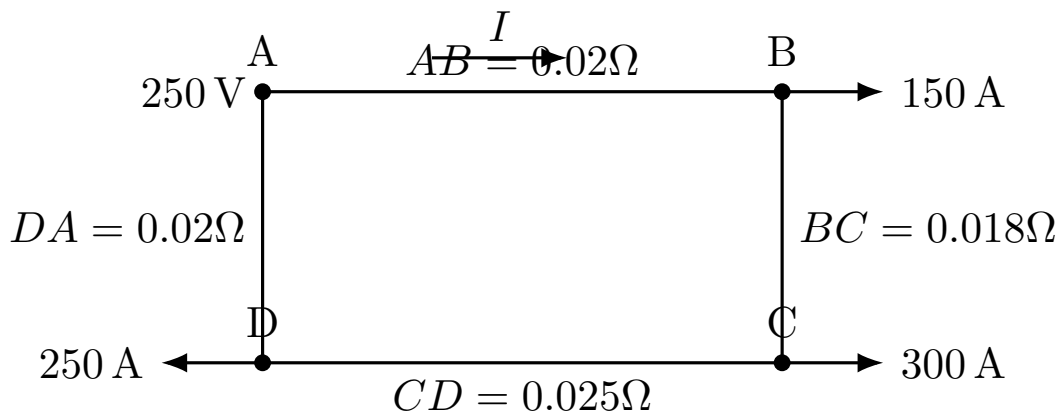


Figure 7: Ring main distributor without interconnector

Let current in section AB be I clockwise.

Then section currents are:

$$I_{AB} = I,$$

$$I_{BC} = I - 150,$$

$$I_{CD} = I - 150 - 300 = I - 450,$$

$$I_{DA} = I - 150 - 300 - 250 = I - 700.$$

For a closed ring,

$$\sum IR = 0.$$

$$0.02I + 0.018(I - 150) + 0.025(I - 450) + 0.02(I - 700) = 0.$$

Solving,

$$0.083I - 27.95 = 0.$$

$$I = 336.75 \text{ A.}$$

Thus,

$$I_{AB} = 336.75 \text{ A,}$$

$$I_{BC} = 186.75 \text{ A,}$$

$$I_{CD} = -113.25 \text{ A,}$$

$$I_{DA} = -363.25 \text{ A.}$$

Negative sign means actual current direction is opposite to assumed clockwise direction.

Step 2: Voltages at B, C and D

At A, voltage is 250 V.

Voltage drop in AB:

$$\Delta V_{AB} = 336.75 \times 0.02 = 6.735 \text{ V.}$$

$$V_B = 250 - 6.735 = 243.27 \text{ V.}$$

Voltage drop in BC:

$$\Delta V_{BC} = 186.75 \times 0.018 = 3.3615 \text{ V.}$$

$$V_C = 243.27 - 3.3615 = 239.90 \text{ V.}$$

Voltage drop in CD:

$$\Delta V_{CD} = (-113.25) \times 0.025 = -2.831 \text{ V.}$$

$$V_D = 239.90 - (-2.831) = 242.73 \text{ V.}$$

Final Answer

Without interconnector:

$$\boxed{V_B = 243.27 \text{ V}}, \quad \boxed{V_C = 239.90 \text{ V}}, \quad \boxed{V_D = 242.73 \text{ V}}.$$

Step 3: With interconnector between A and C

Now A and C are linked by an interconnector of resistance 0.02Ω .

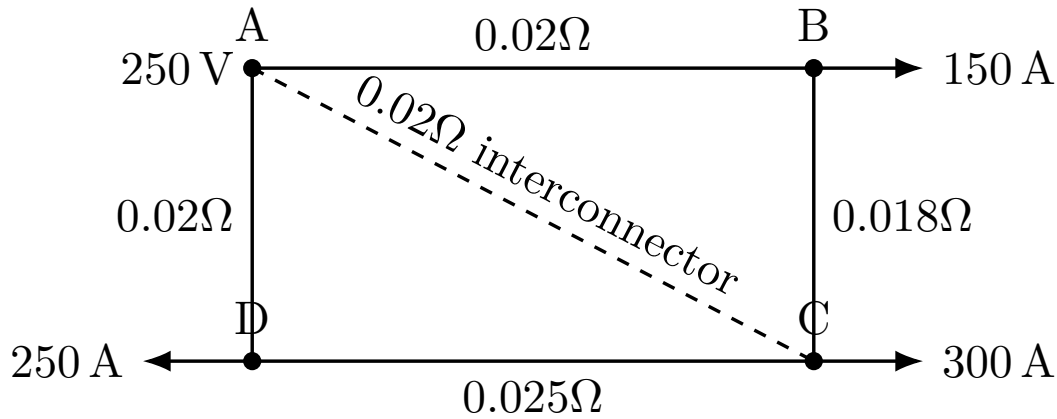


Figure 8: Ring main distributor with A-C interconnector

Use node-voltage method. Let voltages at B , C and D be V_B , V_C , V_D . Since A is directly fed,

$$V_A = 250 \text{ V}.$$

KCL at B :

$$\frac{V_B - 250}{0.02} + \frac{V_B - V_C}{0.018} + 150 = 0.$$

KCL at C :

$$\frac{V_C - V_B}{0.018} + \frac{V_C - V_D}{0.025} + \frac{V_C - 250}{0.02} + 300 = 0.$$

KCL at D :

$$\frac{V_D - V_C}{0.025} + \frac{V_D - 250}{0.02} + 250 = 0.$$

Solving these three simultaneous equations,

$$V_B = 245.961 \text{ V},$$

$$V_C = 245.027 \text{ V},$$

$$V_D = 245.012 \text{ V}.$$

Final Answer

With A-C interconnector:

$$\boxed{V_B = 245.96 \text{ V}}, \quad \boxed{V_C = 245.03 \text{ V}}, \quad \boxed{V_D = 245.01 \text{ V}}.$$

The interconnector improves the voltage at the low-voltage point C .

4.2 (ii) Efficiency increases with transmission voltage

For a three-phase transmission line,

$$P = \sqrt{3}VI \cos \phi.$$

For a fixed transmitted power P and fixed power factor,

$$I = \frac{P}{\sqrt{3}V \cos \phi}.$$

So if voltage V increases, current I decreases.

Line loss is

$$P_{loss} = 3I^2 R.$$

Substituting current,

$$P_{loss} = 3 \left(\frac{P}{\sqrt{3}V \cos \phi} \right)^2 R.$$

$$\boxed{P_{loss} = \frac{P^2 R}{V^2 \cos^2 \phi}}.$$

Hence,

$$P_{loss} \propto \frac{1}{V^2}.$$

Transmission efficiency is

$$\eta = \frac{P}{P + P_{loss}} \times 100\%.$$

As V increases, P_{loss} decreases. Therefore, denominator $P + P_{loss}$ decreases and efficiency increases.

Final Answer

Higher transmission voltage $\Rightarrow$ lower current $\Rightarrow$ lower $I^2 R$ loss $\Rightarrow$ higher efficiency.

Question 5 Group A, Question 5

Question

- (i) Write the main advantages of AC transmission over DC transmission.
 (ii) A DC distributor shown in Fig. Q5(ii) is fed at both ends with 200 V. The resistance of the distributor, including go and return, is $0.001\Omega/\text{m}$. Determine the point of minimum voltage and the value of minimum voltage.

5.1 (i) Advantages of AC transmission over DC transmission

1. **Voltage transformation is easy:** AC voltage can be stepped up or stepped down easily by transformers.
2. **Generation at high voltage is possible:** Alternators can generate AC power at high voltage.
3. **Economical long-distance transmission:** By stepping up voltage, current becomes small, so copper loss and voltage drop reduce.
4. **Substations are simpler:** AC substations are generally cheaper and easier than DC converter stations.
5. **AC machines are simple and robust:** Transformers and induction motors work naturally on AC supply.
6. **Protection and switching are convenient:** Current naturally becomes zero twice in every cycle, which helps arc extinction in AC circuit breakers.

Final Answer

The main advantage of AC transmission is that voltage can be changed easily by using transformers. So high-voltage transmission becomes simple and economical.

5.2 (ii) DC distributor fed at both ends

Diagram and data

The distributor is fed at both ends *A* and *B* at 200 V.

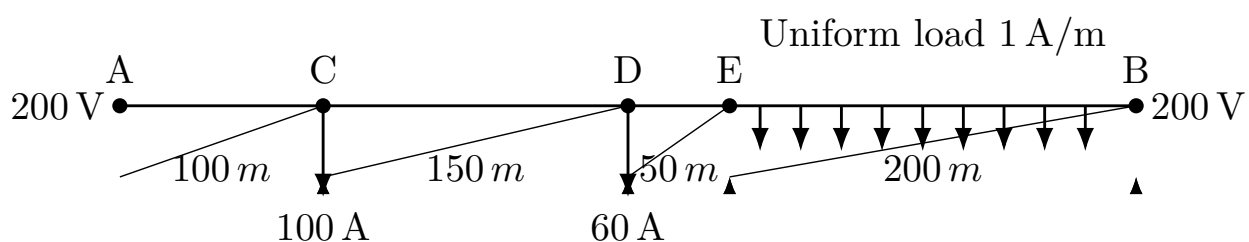


Figure 9: DC distributor fed at both ends

Given loop resistance per metre:

$$r = 0.001\Omega/\text{m}.$$

Assume current supplied from A

Let current supplied from A be I_A .

Section currents:

$$I_{AC} = I_A.$$

After 100 A load at C,

$$I_{CD} = I_A - 100.$$

After 60 A load at D,

$$I_{DE} = I_A - 160.$$

In section EB , uniform loading is 1 A/m over 200 m. At a distance x from E towards B ,

$$I_x = I_A - 160 - x.$$

Since both ends are at equal voltage,

$$\text{Total drop from A to B} = 0.$$

$$0.001 \left[I_A(100) + (I_A - 100)(150) + (I_A - 160)(50) + \int_0^{200} (I_A - 160 - x)dx \right] = 0.$$

Now,

$$\int_0^{200} (I_A - 160 - x)dx = 200I_A - 32000 - \frac{200^2}{2}.$$

$$= 200I_A - 52000.$$

So,

$$100I_A + 150I_A - 15000 + 50I_A - 8000 + 200I_A - 52000 = 0.$$

$$500I_A - 75000 = 0.$$

$$I_A = 150 \text{ A}.$$

Therefore,

$$I_{AC} = 150 \text{ A},$$

$$I_{CD} = 50 \text{ A},$$

$$I_{DE} = 150 - 160 = -10 \text{ A}.$$

Current changes direction at point D . Therefore, the point of minimum voltage is D .

Minimum voltage

Voltage drop from A to D is

$$\Delta V_{AD} = 150(100)(0.001) + 50(150)(0.001).$$

$$\Delta V_{AD} = 15 + 7.5 = 22.5 \text{ V}.$$

$$V_D = 200 - 22.5 = 177.5 \text{ V}.$$

Final Answer

Point of minimum voltage = D

$$V_{min} = 177.5 \text{ V}$$

Question 6 Group B, Question 6

Question

- (i) Why are circuit breakers and isolators used in substations?
- (ii) Explain the operation of relay in power system.
- (iii) Mention the factors on which earth resistance depends.

6.1 (i) Use of circuit breakers and isolators in substations

Circuit breaker

A **circuit breaker** is a switching device which can make or break a circuit under normal condition as well as under fault condition.

Isolator

An **isolator** is a mechanical switch used to isolate a part of the circuit after the circuit breaker has opened the circuit. It is operated only under no-load condition.

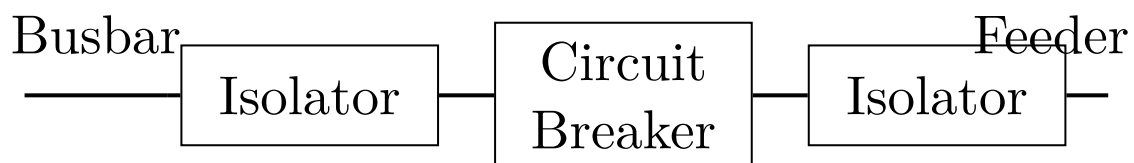


Figure 10: Circuit breaker with isolators in a feeder bay

Why both are required

- Circuit breaker interrupts fault current safely.
- Isolator gives visible isolation for maintenance.
- Circuit breaker is opened first, then isolator is opened.
- During closing, isolator is closed first, then circuit breaker is closed.
- Isolator should not be opened on load because it cannot safely interrupt heavy current.

Final Answer

Circuit breaker is used for fault interruption. Isolator is used for safe isolation after interruption. Both are required for safe operation and maintenance of a substation.

6.2 (ii) Operation of relay in power system

Definition

A **protective relay** is a device which senses abnormal conditions like overcurrent, short circuit or earth fault and gives a trip signal to the circuit breaker.

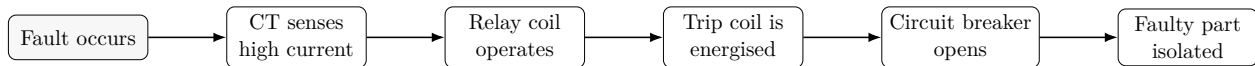
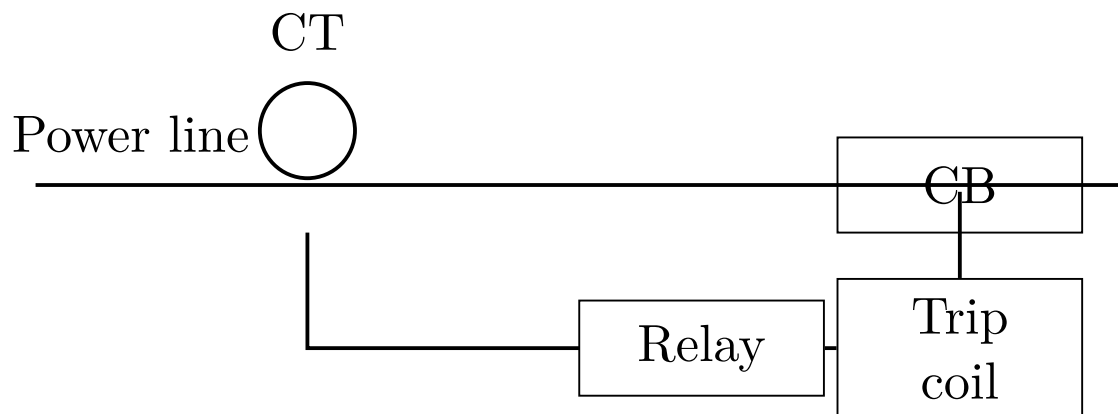


Figure 11: Flow of relay operation



Secondary current operates relay.
Relay closes trip circuit of CB.

Figure 12: Simple relay and circuit breaker operating diagram

Working steps

1. During normal condition, relay remains inactive.
2. When a fault occurs, current increases suddenly.
3. Current transformer sends a proportional current to the relay.
4. Relay detects that current has crossed the set value.
5. Relay contacts close the trip circuit.
6. Trip coil opens the circuit breaker.
7. Faulty section is disconnected from the healthy system.

Final Answer

A relay does not directly break the power circuit. It only detects the fault and commands the circuit breaker to open.

6.3 (iii) Factors affecting earth resistance

Earth resistance depends on the following factors:

1. Type and material of earth electrode.
2. Size of electrode and earth wire.
3. Moisture content of soil.
4. Temperature of soil.
5. Depth of electrode below ground level.
6. Salt and charcoal treatment around the electrode.
7. Number of electrodes connected in parallel.
8. Nature of soil, such as rocky, sandy or clay soil.

Final Answer

Earth resistance should be low. Moist soil, deep electrode and proper salt-charcoal treatment help to reduce earth resistance.

Question 7 Group B, Question 7

Question

- (i) What do you understand by stranded conductor? What are its advantages?
- (ii) Discuss the major tasks performed by a substation.
- (iii) What do you understand by double busbar arrangement in a substation?

7.1 (i) Stranded conductor

Definition

A **stranded conductor** is made by twisting many small wires, called strands, together. It is used instead of a single solid conductor to improve flexibility and mechanical strength.

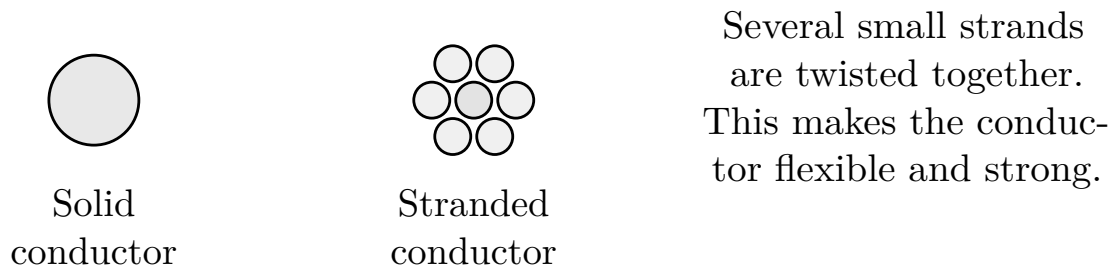


Figure 13: Solid and stranded conductors

Important points

- A stranded conductor is flexible and can be coiled easily.
- It is suitable for long transmission and distribution lines.
- Strands are arranged in layers.
- Alternate layers are usually twisted in opposite directions for mechanical stability.
- A notation like 19/0.64 means 19 strands, each of diameter 0.64 mm.

For a stranded conductor,

$$D = (2x - 1)d,$$

where D is the overall diameter, x is the number of layers and d is the diameter of each strand.

Advantages

1. It has more flexibility than a solid conductor.
2. It can be transported and erected easily.
3. It gives better mechanical strength.

4. It is less likely to break due to vibration and wind.
5. It permits use of steel core in ACSR conductors for long spans.

Final Answer

A stranded conductor is preferred in overhead lines because it is flexible, strong and easy to handle during erection.

7.2 (ii) Major tasks performed by a substation

Definition

A **substation** is an assembly of electrical equipment used to control, transform, protect, measure and distribute electrical power.

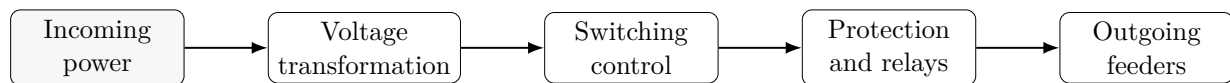


Figure 14: Main tasks of a substation

Major tasks

1. **Voltage transformation:** It steps up or steps down voltage using transformers.
2. **Switching operation:** It connects or disconnects lines and equipment.
3. **Protection:** It isolates faulty sections using relays and circuit breakers.
4. **Voltage regulation:** It helps maintain voltage within required limits.
5. **Power factor improvement:** Capacitor banks or other devices may be used.
6. **Metering and measurement:** Voltage, current, power and energy are measured.
7. **Distribution of power:** It sends power to different feeders and consumers.
8. **Earthing and safety:** It provides safe operation during fault and lightning conditions.

Final Answer

A substation is not only a transformer point. It is also a control, protection, measurement and distribution point in the power system.

7.3 (iii) Double busbar arrangement

Definition

In a **double busbar arrangement**, two busbars are provided in the substation. A feeder or transformer circuit can be connected to either busbar through isolators and a circuit breaker. A bus coupler is used to connect the two buses when required.

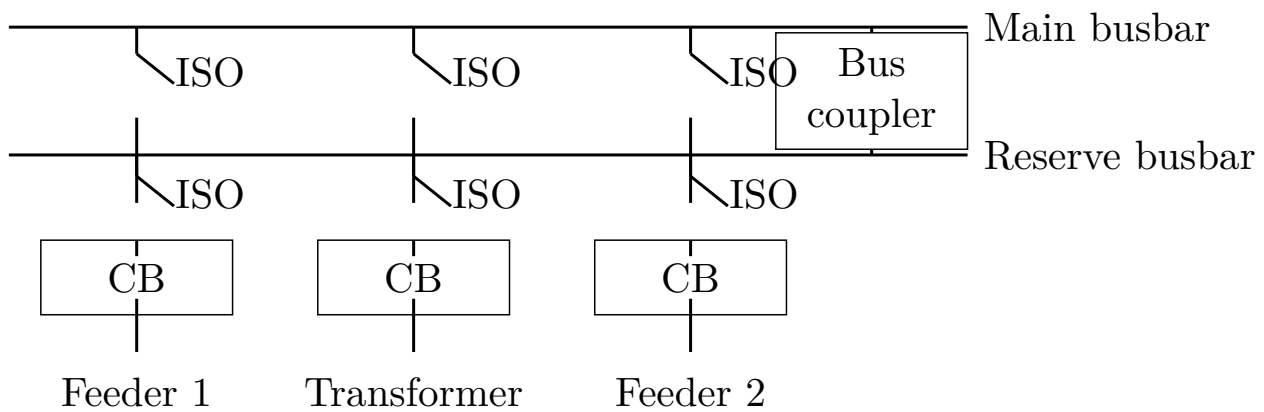


Figure 15: Simple double busbar arrangement

Working idea

Normally, one busbar carries the load and the other remains as reserve. During maintenance or fault, the load can be transferred from one busbar to another by using isolators and the bus coupler.

Advantages

1. It gives better reliability than single busbar.
2. Maintenance of one busbar is possible without complete shutdown.
3. Load can be transferred from one busbar to the other.
4. It gives flexibility in operation.

Disadvantages

1. Cost is higher than single busbar arrangement.
2. More isolators and equipment are required.
3. Protection and operation are more complicated.

Final Answer

Double busbar arrangement uses two busbars to improve reliability and flexibility. It is suitable for important substations where interruption of supply should be avoided.

Final Exam Revision Table

Topic	Main formula / idea	Exam remark
Loop distribution	Supply from two sides	Better continuity and voltage regulation
Feeder	Designed by current capacity	No tapping usually
Distributor	Designed by voltage drop	Consumer tappings taken
Uniform distributor fed one end	$v_x = ir(lx - x^2/2)$	Minimum voltage at far end
Conductor comparison	Volume = nal	Always state assumptions
Kelvin's law	$P_2a = P_3/a$	Variable capital charge equals loss cost
Ring main	$\sum IR = 0$	Treat as closed-loop KVL problem
Transmission efficiency	$P_{loss} = P^2R/(V^2 \cos^2 \phi)$	Higher voltage gives lower loss
Distributor fed both ends	Current reversal point	Minimum voltage occurs there
Relay	Senses fault and trips CB	Relay does not break main circuit directly
Stranded conductor	$D = (2x - 1)d$	Flexible and mechanically strong
Substation	Transform, switch, protect, meter	Control point of power system
Double busbar	Two busbars with bus coupler	Reliable but costlier

End of Part-II Complete Solutions